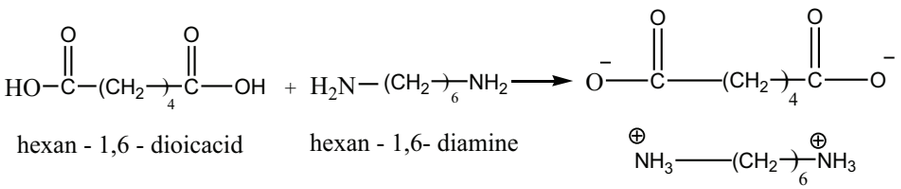
Condensation polymerisation

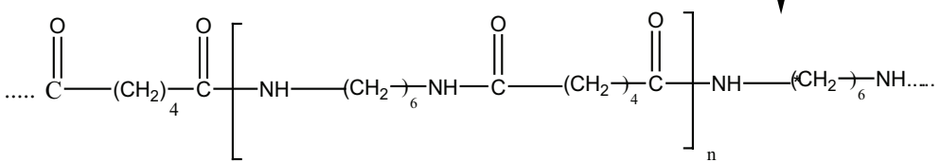
Condensation polymers are formed by the reaction between functional groups an adjacent monomers with the elimination of simple molecules like H_2O, NH_3 etc…. Each monomer must undergo at least two substitution reactions to continue to grow the polymer chain i.e., the monomer must be at least bi functional. Examples : Nylon– 6,6, terylene….

Nylon – 6,6

Nylon – 6,6 can be prepared by mixing equimolar adipic acid and hexamethylene – diamine to form a nylon salt which on heating eliminate a water molecule to form amide bonds.



hexan - 1,6 - dioicacid hexan - 1,6- diamine Nylon salt

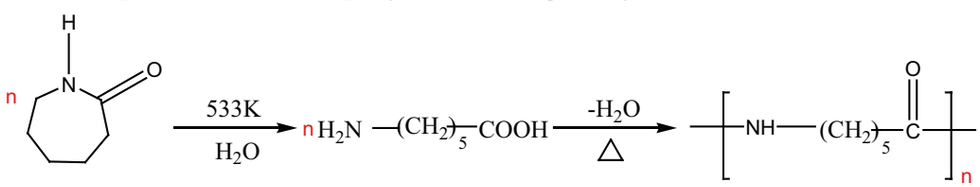


Poly (hexamethyleneadipamide)
Nylon 6,6

It is used in textiles, manufacture of cards etc…

Nylon – 6

Capro lactam (monomer) on heating at 533K in an inert atmosphere with traces of water gives ϵ- amino carproic acid which polymerises to give nylon – 6



Nylon -6

It is used in the manufacture of tyrecards fabrics etc….

II. Preparation of terylene (Dacron)

The monomers are ethylene glycol and terepathalic acid (or) dimethylterephthalate. When these monomers are mixed and heated at 500K in the presence of zinc acetate and antimony trioxide catalyst, terylene is formed.

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